

STAT 516 hw 2

AUTHOR

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Several statistics students recorded their heights and the lengths of their index fingers. The measurements, in centimeters and millimeters, respectively, are read into R using the code below (copy and paste this into your .qmd file):

```
link <- url("https://gregorkb.github.io/data/height_length_index_pinky_shoesize_combined")
hl <- read.table(link, sep=",", header=T)
head(hl)
Y <- (hl$ft*12 + hl$in.)*2.54 # heights in cm
x <- hl$ind # index finger lengths in mm
n <- length(Y)
```

It is of interest to use the simple linear regression model to predict the height of a person based on the length of his or her index finger.

1)

Make a scatterplot of the heights versus the index finger lengths with the least-squares line overlaid. Report the intercept and slope of the least-squares line as well as the value of Pearson's correlation coefficient.

2)

Based on the fitted simple linear regression model, to what difference in height does an additional millimeter of index finger length correspond?

3)

Give an estimate of the error term variance.

4)

Make a normal quantile-quantile plot of the residuals as well as a residuals versus fitted values plot. Then carefully explain whether you believe the assumptions of the simple linear regression model are satisfied for these data.

5)

Give a confidence interval for the slope parameter .

6)

State whether you would reject : versus : at the significance level.

7)

Give an estimate of the mean height of persons with index finger length equal to mm. In addition, give a 95% confidence interval for this mean height.

8)

A hand print is found made by a hand with an index finger length of mm. Give an interval which will contain with probability the height of the person who made the print.

9)

Do you think the above interval would be useful in identifying the person who made the print?

10)

Give the value of the coefficient of determination for these data. Interpret the value.

11)

Give the value of the test statistic _____ as well as the value of _____.

12)

Give the p-value for testing : versus : based on the value of the test statistic . Interpret your answer.

13)

Comment on whether there are any outliers in the data set. Show a plot to support your answer.

14)

Convert the heights to inches (divide by) and fit the regression model again. Report the coefficient of determination as well as the p-value for testing whether the slope coefficient is equal to zero. Compare these to the values you obtained previously.